

# FEEDBACK MANAGEMENT SYSTEM

By Anisha Sinha, Archit Agarwal, Mayank Chopra

GitHub Link: [github.com/AnieSinha/feedback-management-system](https://github.com/AnieSinha/feedback-management-system)

Deployed link: [feedback-management-ongh.onrender.com](https://feedback-management-ongh.onrender.com)

## 1. Project Overview

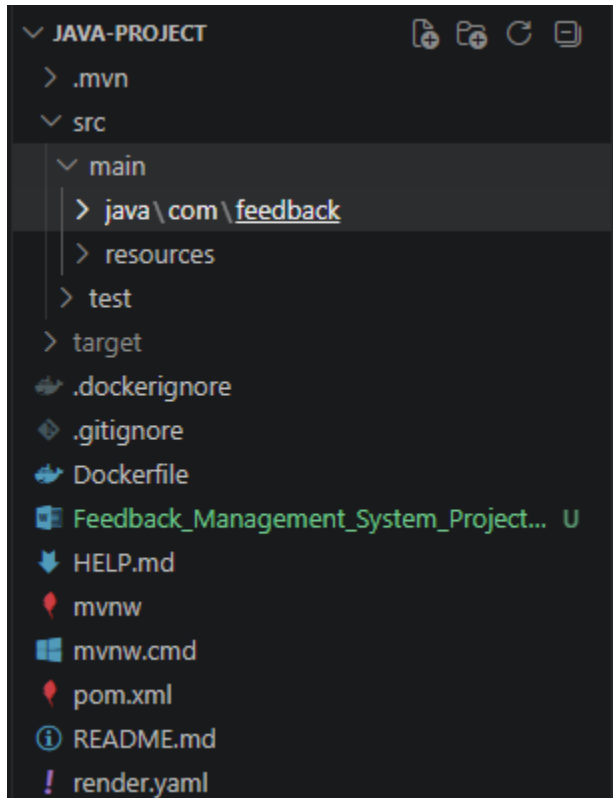
The Feedback Management System is an end-to-end training feedback platform that lets an organisation manage faculty, courses, and training programs, collect structured feedback from participants, and turn that feedback into actionable reports. It is built as a Spring Boot REST API backed by MongoDB, secured with JWT authentication and role-based access control across three roles: Admin, Coordinator, and Participant. Admins manage faculty, their skills, and courses; coordinators create and run training programs and enrol participants; participants submit and update feedback on the programs they attend. The system is containerised with Docker and deployed on Render against a managed MongoDB Atlas cluster.

### 1.1 User Stories

| Actor          | User Story                                                                                                                                                                                                                                                  |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Anisha Sinha   | As a developer, I want to build the authentication and faculty management layer with features: user login, role-based access control, faculty create, update, and delete operations, and faculty skill management                                           |
| Archit Agarwal | As a developer, I want to build the course and training program management layer: course create, update, delete, and listing, along with training program creation, modification, deletion, viewing, and participant enrolment.                             |
| Mayank Chopra  | As a developer, I want to build the feedback and reporting layer: feedback submission and editing, feedback reports, defaulter tracking, training summary reports, and course feedback analytics and connect the system to MongoDB and deploy it on Render. |

## 2. Technical Architecture

### 2.1 Project Structure



### 2.2 Core Services

The system is organised into layered Spring components backed by a single MongoDB database.

| Component                 | Description                                                                                                                          |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| REST controllers          | Spring Boot @RestController layer exposing the auth, faculty, skill, course, training program, feedback, report, and user endpoints. |
| Service layer             | Business logic for each module, including feedback validation, duplicate-submission checks, and report aggregation.                  |
| Spring Security + JWT     | Stateless authentication. A JWT filter validates the bearer token on each request and populates the security context.                |
| Role-based access control | Method-level @PreAuthorize rules across three roles: ADMIN, COORDINATOR, and PARTICIPANT.                                            |

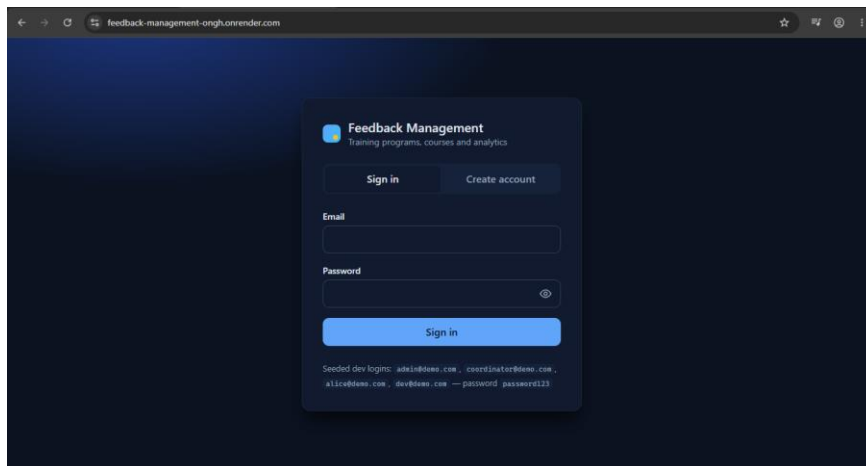
|                       |                                                                                                                                |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|
| MongoDB (Spring Data) | Document store for users, faculty, skills, courses, training programs, and feedback. Repository interfaces derive the queries. |
| MongoDB Atlas         | Managed cloud cluster used by the deployed application, configured through the MONGODB_URI environment variable.               |
| Static dashboard      | HTML, CSS, and JavaScript front-end served by Spring Boot that consumes the REST API and renders charts.                       |
| Docker + Render       | Docker image deployed as a Render web service, configured by environment variables.                                            |

**User-1: Anisha Sinha:** As a developer, I want to build authentication and faculty management, so that only authorised users can access the system and faculty records and skills are maintained accurately.

## 1. User Login

Scenario: Training admins, coordinators, and participants log in with their email and password, and receive a JWT that authorises subsequent requests. Valid users reach the application; invalid credentials are rejected.

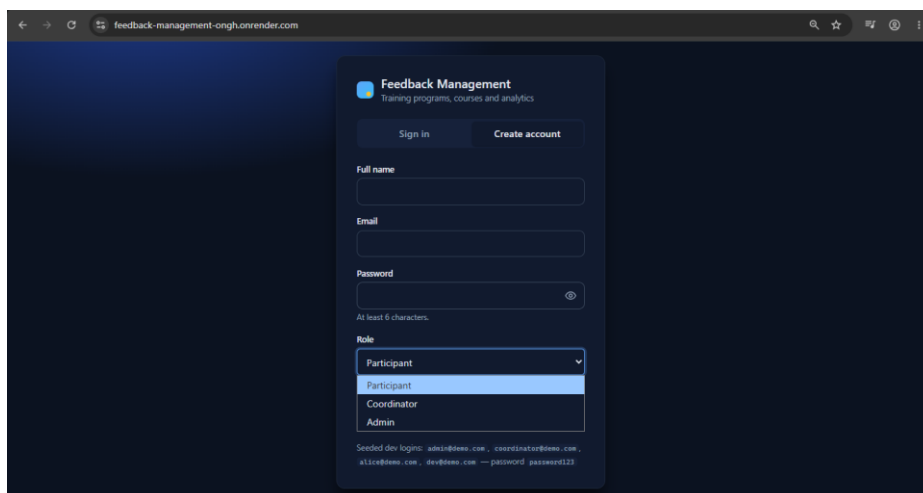
Output:



## 2. Role-Based Access

Scenario: Each user sees and can act on only the features allowed by their role. Method-level authorisation rules enforce the correct permissions for Admin, Coordinator, and Participant.

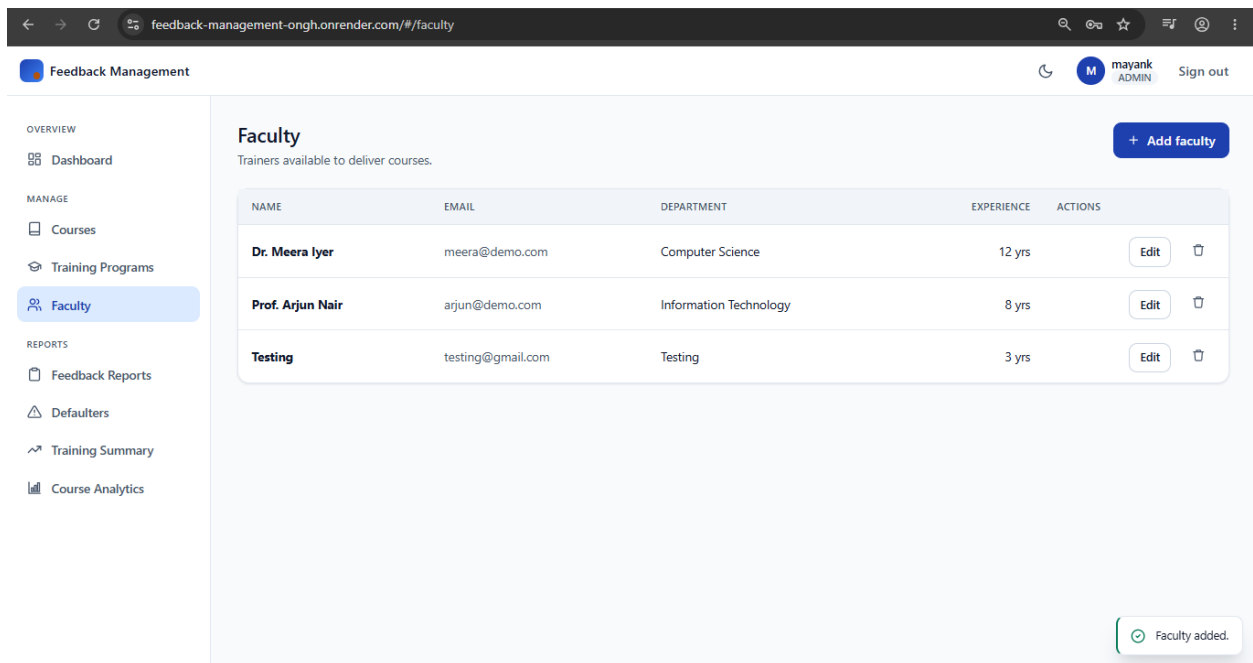
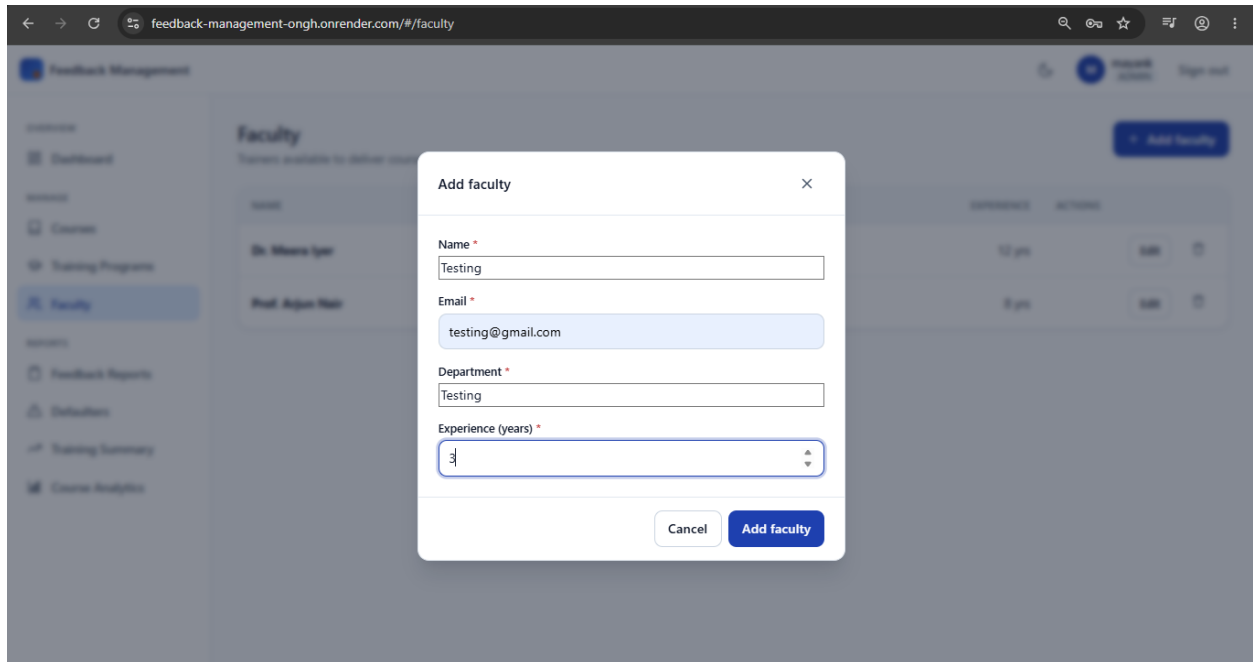
Output:



### 3. Add Faculty

Scenario: An admin adds a faculty member with name, email, department, and years of experience, and the record is saved successfully.

Output:



## 4. Update Faculty

Scenario: An admin edits an existing faculty member's details and the updated information is persisted.

Output:

The screenshot shows a web application titled "Feedback Management" with a sidebar menu. The main content area is titled "Faculty" and displays a table of faculty members. A modal window titled "Edit faculty" is open, allowing an admin to update a faculty member's details. The modal contains the following fields:

- Name \*: Mayank
- Email \*: testing@gmail.com
- Department \*: Testing
- Experience (years) \*: 3

At the bottom of the modal are two buttons: "Cancel" and "Save changes".

## 5. Delete Faculty

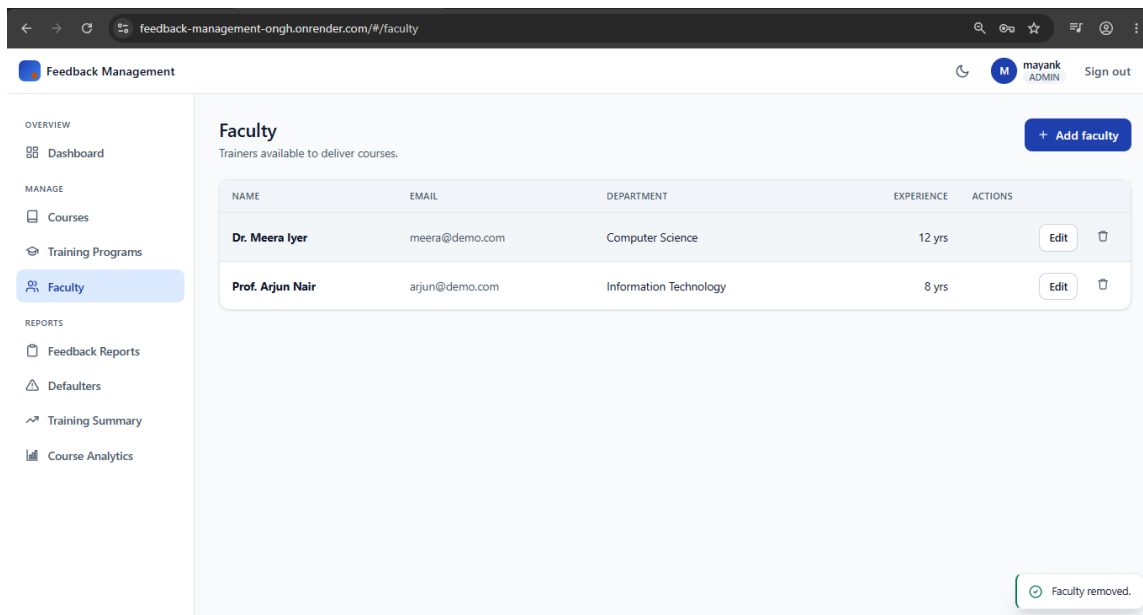
Scenario: An admin removes a faculty member and the record is deleted successfully.

Output:

The screenshot shows the same web application as before, but now a modal window titled "Delete faculty" is open. The modal contains the following text:

Remove Mayank from the faculty list?

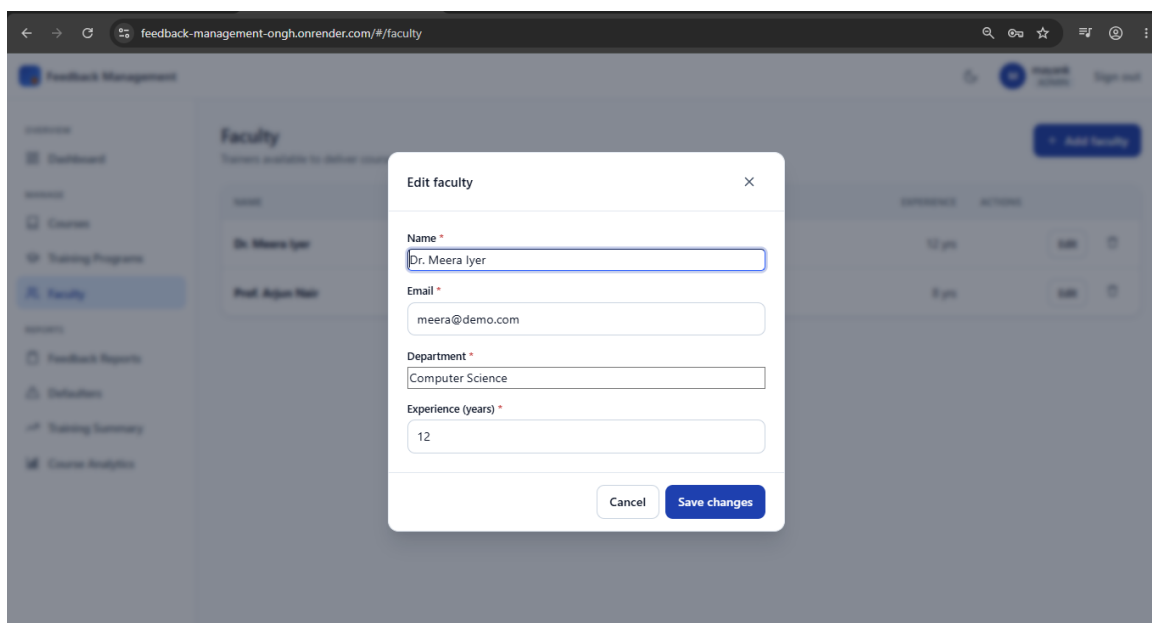
At the bottom of the modal are two buttons: "Cancel" and "Delete".



## 6. Maintain Faculty Skills

Scenario: An admin manages the skill set mapped to each faculty member, adding, updating, and removing skills so that expertise is recorded correctly.

Output:



## 7. Add Course

Scenario: An admin adds a training course with a name, description, and duration, and the course is created successfully.

[illegible]

Deliver secure authentication and authorisation, and a reliable faculty management module that keeps faculty records and their skills accurate.

- Implement user registration and JWT-based login
- Enforce role-based authorisation for Admin, Coordinator, and Participant
- Build the faculty entity, repository, and REST API
- Add update and delete operations for faculty records
- Create the skill management module mapped to faculty
- Return consistent error responses for invalid credentials and access

- Only authenticated users can reach protected endpoints
- Each role is restricted to the actions it is permitted to perform
- Faculty records can be created, updated, and deleted reliably
- Faculty skills are mapped and maintained accurately



The screenshot shows the 'Feedback Management' dashboard on the OnRender platform. The 'Courses' section is active, displaying a list of courses. A modal dialog titled 'Delete course' is open, asking for confirmation to delete the course 'Java-Testing?'. The dialog includes a message: 'Delete "Java-Testing"? Feedback already submitted for it will remain in the database.' and two buttons: 'Cancel' and 'Delete'.

| COURSE                   | DESCRIPTION | Duration | ACTIVATED |
|--------------------------|-------------|----------|-----------|
| Spring Boot Fundamentals |             | 40 hrs   | ON        |
| Core Java                |             | 90 hrs   | ON        |
| MongoDB Essentials       |             | 24 hrs   | ON        |
| Java Testing             |             | 30 hrs   | ON        |

Feedback Management

OVERVIEW

Dashboard

MANAGE

Courses

Training Programs

Faculty

REPORTS

Feedback Reports

Defaulters

Training Summary

Course Analytics

Courses

Courses that participants can be asked to give feedback on.

+ Add course

| COURSE                   | DESCRIPTION                                  | DURATION | ACTIONS                     |
|--------------------------|----------------------------------------------|----------|-----------------------------|
| Spring Boot Fundamentals | REST APIs, dependency injection, data access | 40 hrs   | <div>Edit</div> <div></div> |
| Core Java                | OOP, collections, streams, concurrency       | 60 hrs   | <div>Edit</div> <div></div> |
| MongoDB Essentials       | Documents, indexes, aggregation pipelines    | 24 hrs   | <div>Edit</div> <div></div> |

Course deleted.

### 3. View Courses

Scenario: Any authenticated user can view the list of available courses, and the course list is displayed.

Output:

Feedback Management

OVERVIEW

Dashboard

MANAGE

Courses

Training Programs

FEEDBACK

Submit Feedback

My Feedback

Welcome, Alice Kumar

Your enrolled programs and the feedback you have shared.

Enrolled programs

2

you can submit feedback for these

Feedback submitted

3

thank you

Awaiting your feedback

3

course and program pairs

Your average rating

5.0

across your submissions

Your programs

+ Submit feedback

| PROGRAM                                                                    | STATUS   | STARTS       | ENDS         |
|----------------------------------------------------------------------------|----------|--------------|--------------|
| Java Backend Bootcamp<br>Twelve week intensive backend engineering program | ONGOING  | Jun 1, 2026  | Aug 24, 2026 |
| Data Engineering Track<br>Databases and pipelines for backend engineers    | UPCOMING | Jul 27, 2026 | Oct 19, 2026 |

Feedback Management

OVERVIEW

Dashboard

MANAGE

Courses

Training Programs

FEEDBACK

Submit Feedback

My Feedback

Alice Kumar

PARTICIPANT

Sign out

Courses

Courses that participants can be asked to give feedback on.

| COURSE                   | DESCRIPTION                                  | DURATION |
|--------------------------|----------------------------------------------|----------|
| Spring Boot Fundamentals | REST APIs, dependency injection, data access | 40 hrs   |
| Core Java                | OOP, collections, streams, concurrency       | 60 hrs   |
| MongoDB Essentials       | Documents, indexes, aggregation pipelines    | 24 hrs   |

## 4. Create Training Program

Scenario: A coordinator creates a training program with a name, schedule, participant limit, and the courses it covers, and the program is created successfully.

Output:

Feedback Management

OVERVIEW

Dashboard

MANAGE

Courses

Training Programs

REPORTS

Feedback Reports

Defaulters

coordinator

COORDINATOR

Sign out

Dashboard

Programs you coordinate and the feedback they have attracted.

Training programs

2

assigned to you

Enrolled participants

6

across those programs

Feedback received

9

total submissions

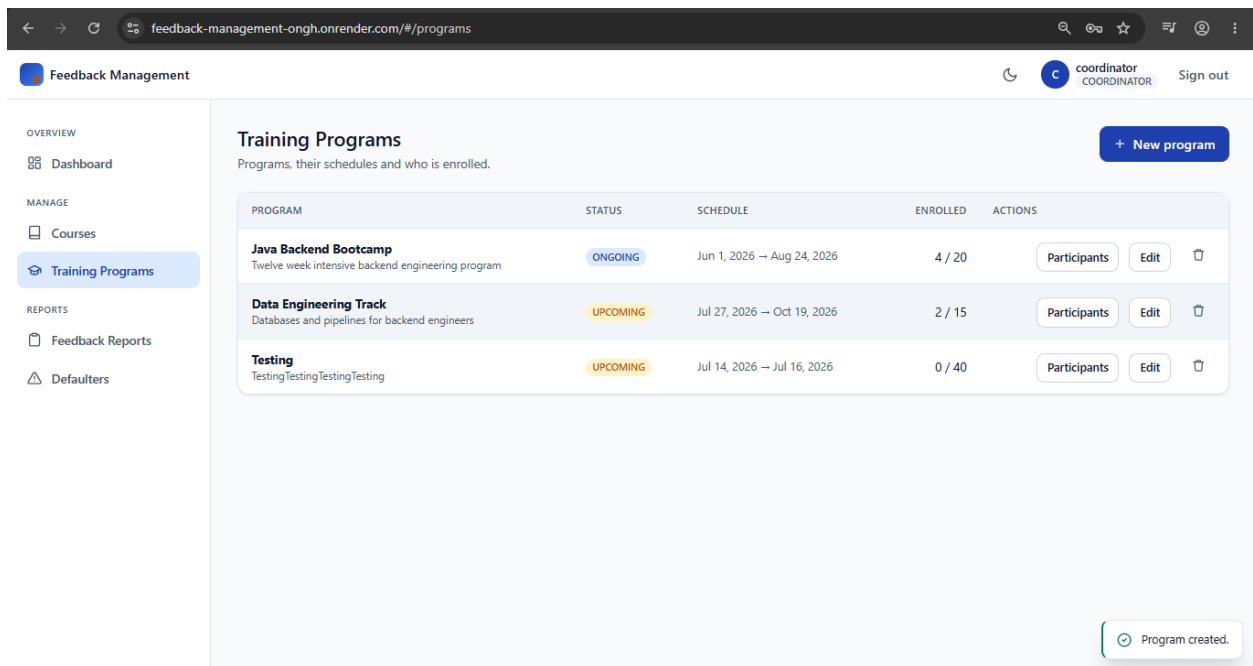
Average rating

3.89

across all feedback

Your programs

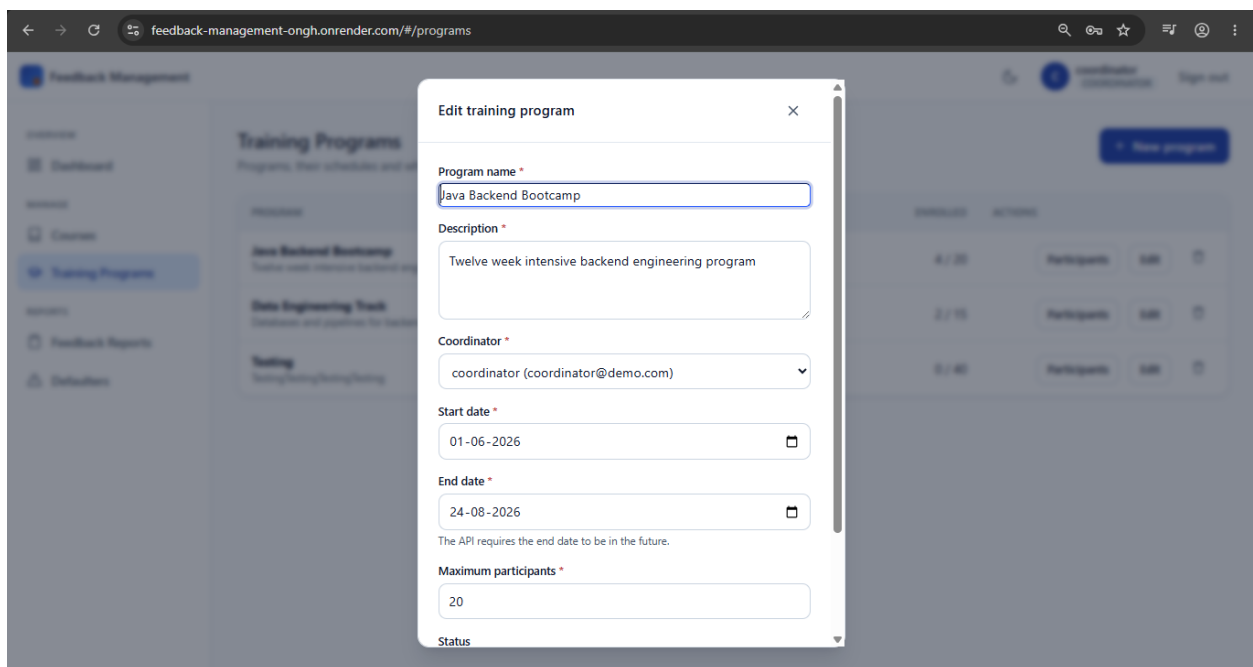
| PROGRAM                | STATUS   | ENROLLED | ENDS         | DEFAULTERS                  |
|------------------------|----------|----------|--------------|-----------------------------|
| Java Backend Bootcamp  | ONGOING  | 4 / 20   | Aug 24, 2026 | <a href="#">View report</a> |
| Data Engineering Track | UPCOMING | 2 / 15   | Oct 19, 2026 | <a href="#">View report</a> |



## 5. Modify Training Program

Scenario: A coordinator modifies an existing training program and the changes are saved successfully.

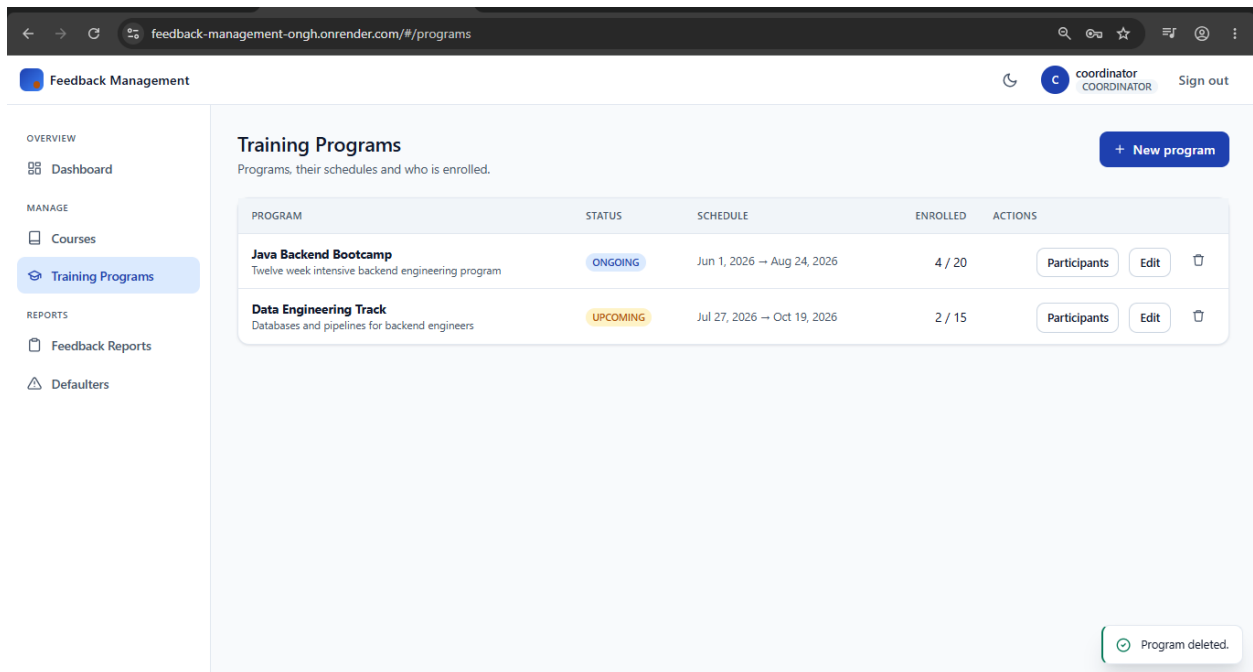
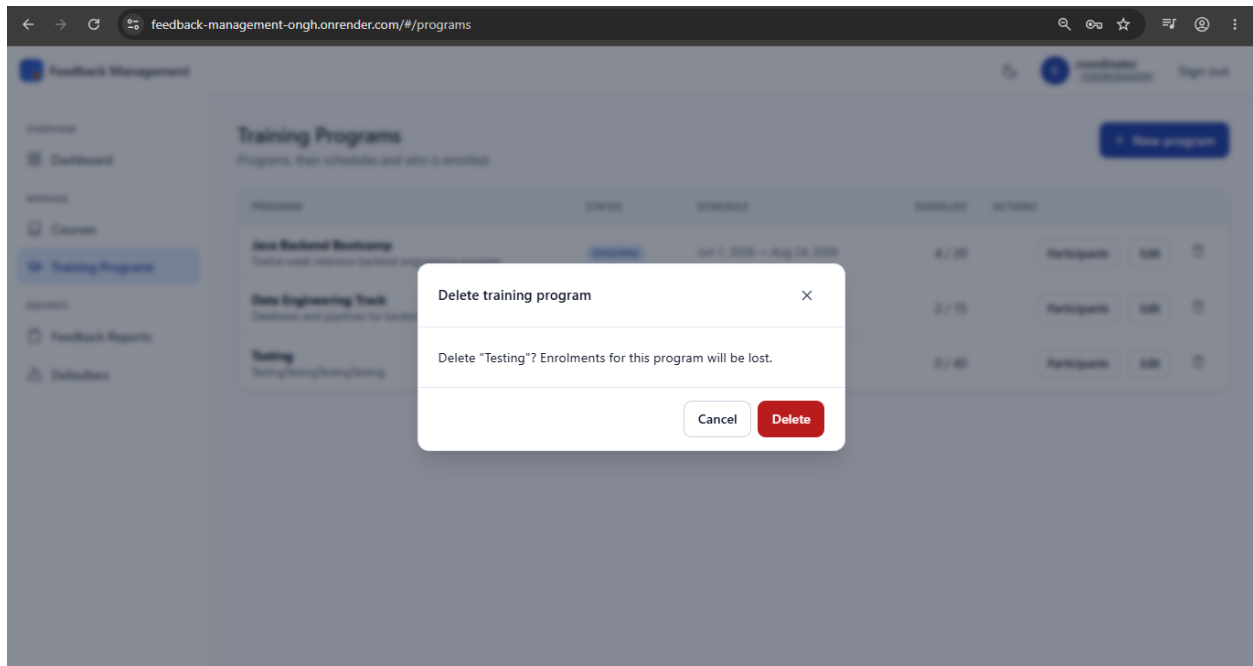
Output:



## 6. Delete Training Program

Scenario: A coordinator deletes a training program and it is removed successfully.

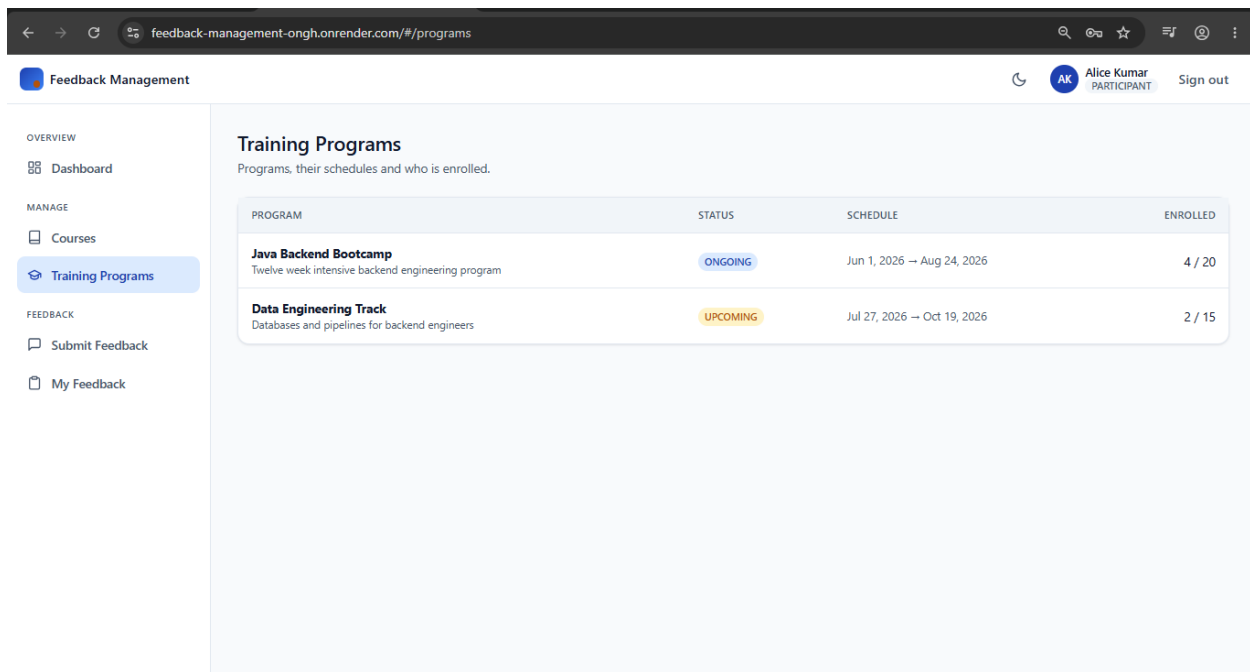
Output:



## 7. View Training Programs

Scenario: Users can view training programs and their details, including enrolled participants.

Output:



## Objective

Deliver a complete, well-authorised catalogue of courses and training programs that the feedback and reporting modules build on.

## Key Analysis

- Build the course entity, repository, and CRUD API
- Restrict course creation, updates, and deletion to admins
- Build the training program entity and its CRUD API
- Allow coordinators to enrol and remove participants
- Expose program listings, including lookup by coordinator
- Validate program data such as schedule and participant limits

## Outcomes

- Admins can maintain the course catalogue end to end
- Coordinators can create, modify, and delete training programs
- Participants can be enrolled in and removed from programs
- Course and program listings are available to the whole application

**User-3: Mayank Chopra:** As a developer, I want to build the feedback and reporting layer and take the system to production, so that feedback is captured, analysed, and available on a live URL.

## 1. Submit Feedback

Scenario: A user submits feedback on a training program they are enrolled in, with a rating and comments, and the feedback is saved successfully. Duplicate submissions for the same program are rejected.

Output:

The screenshot shows the 'Submit Feedback' form in the Feedback Management application. The user is Alice Kumar, a participant. The form is titled 'New feedback' and includes a note: 'Rate a course you attended. One response per course, per program.' The form fields are:

- Training program \***: A dropdown menu with 'Java Backend Bootcamp' selected.
- Course \***: A dropdown menu with 'Spring Boot Fundamentals' selected. A warning message below reads: 'You have already reviewed this course for this program. Edit it from My Feedback instead.'
- Rating \***: A star rating system showing 4 out of 5 stars (4 orange stars, 1 grey star) with the text '4 out of 5 — good'.
- Comments**: A text area containing 'This is the testing' with a character count of '19 / 1000 characters'.

A 'Submit feedback' button is at the bottom of the form. To the right, an 'Already submitted' section shows a table of previous submissions:

| COURSE                   | PROGRAM                | RATING |
|--------------------------|------------------------|--------|
| Spring Boot Fundamentals | Java Backend Bootcamp  | 5 ☆☆☆☆ |
| Core Java                | Java Backend Bootcamp  | 5 ☆☆☆☆ |
| MongoDB Essentials       | Data Engineering Track | 5 ☆☆☆☆ |

Below the table, it says: 'To change a response, use My Feedback.'

The screenshot shows the 'My Feedback' page in the Feedback Management application. The user is Alice Kumar, a participant. The page title is 'My Feedback' with a subtitle: 'Everything you have submitted. You can revise your own responses at any time.' A '+ Submit feedback' button is in the top right corner. The page displays a table of submitted feedback:

| COURSE                   | PROGRAM                | RATING | COMMENTS                                    | SUBMITTED              | ACTIONS |
|--------------------------|------------------------|--------|---------------------------------------------|------------------------|---------|
| Spring Boot Fundamentals | Java Backend Bootcamp  | 5 ☆☆☆☆ | Clear explanations and great hands-on labs. | Jul 9, 2026, 6:05 AM   | Edit    |
| Core Java                | Java Backend Bootcamp  | 5 ☆☆☆☆ | The streams module was excellent.           | Jul 9, 2026, 6:05 AM   | Edit    |
| MongoDB Essentials       | Data Engineering Track | 5 ☆☆☆☆ | Best database course I have taken.          | Jul 9, 2026, 6:05 AM   | Edit    |
| MongoDB Essentials       | Java Backend Bootcamp  | 4 ☆☆☆☆ | This is the testing                         | Jul 14, 2026, 10:45 AM | Edit    |

A green notification bubble at the bottom right says: 'Feedback submitted. Thank you!'

## 2. View Feedback Reports

Scenario: An admin or coordinator views the feedback submitted for their programs, and the reports are generated successfully.

Output:

The screenshot displays the 'Feedback Reports' page in the 'Feedback Management' application. The page title is 'Feedback Reports' with a subtitle 'Every response submitted, filterable by program and course.' Below the title, there are two dropdown menus for 'Training program' (set to 'All programs') and 'Course' (set to 'All courses'), followed by an 'Apply filters' button and a 'Reset' link. The main content area shows '8 responses' with an 'Average 3.88 / 5' rating. A table lists the feedback entries with columns for Participant, Course, Program, Rating, Comments, Submitted, and Actions. The table contains six rows of feedback from Alice Kumar, Bob Sharma, and Chitra Rao.

| PARTICIPANT | COURSE                   | PROGRAM               | RATING  | COMMENTS                                      | SUBMITTED             | ACTIONS |
|-------------|--------------------------|-----------------------|---------|-----------------------------------------------|-----------------------|---------|
| Alice Kumar | Spring Boot Fundamentals | Java Backend Bootcamp | 5 ☆☆☆☆☆ | Clear explanations and great hands-on labs.   | Jul 9, 2026, 6:05 AM  | 🗑️      |
| Bob Sharma  | Spring Boot Fundamentals | Java Backend Bootcamp | 4 ☆☆☆☆☆ | Solid content, wanted more time on security.  | Jul 10, 2026, 6:05 AM | 🗑️      |
| Chitra Rao  | Spring Boot Fundamentals | Java Backend Bootcamp | 2 ☆☆☆☆☆ | Paced too fast for beginners.                 | Jul 12, 2026, 6:05 AM | 🗑️      |
| Alice Kumar | Core Java                | Java Backend Bootcamp | 5 ☆☆☆☆☆ | The streams module was excellent.             | Jul 9, 2026, 6:05 AM  | 🗑️      |
| Bob Sharma  | Core Java                | Java Backend Bootcamp | 3 ☆☆☆☆☆ | Good, but the concurrency section was rushed. | Jul 11, 2026, 6:05 AM | 🗑️      |

## 3. View Defaulters Report

Scenario: An admin or coordinator views the participants enrolled in a training program who have not yet submitted feedback, so that follow-up can be targeted.

Output:

The screenshot displays the 'Defaulters Report' page in the 'Feedback Management' application. The page title is 'Defaulters Report' with a subtitle 'Enrolled participants who have not yet submitted any feedback for a program.' Below the title, there is a 'Training program' dropdown menu set to 'Data Engineering Track'. The main content area features four summary cards: 'Enrolled' (2 participants), 'Responded' (1 submitted at least one feedback), 'Defaulters' (1 no feedback submitted), and 'Response rate' (50.0% of enrolled participants). Below these cards, a table titled 'Defaulters in Data Engineering Track' (with a '1 outstanding' badge) lists the participants who have not submitted feedback. The table has columns for #, Participant, and Email.

| # | PARTICIPANT | EMAIL        |
|---|-------------|--------------|
| 1 | Bob Sharma  | bob@demo.com |



## 4. Update Feedback

Scenario: A participant updates their own feedback before the submission deadline, and the revised feedback is saved.

Output:

The screenshot shows the 'Submit Feedback' page in the Feedback Management system. The page has a sidebar with 'Dashboard', 'Courses', 'Training Programs', 'Submit Feedback', and 'My Feedback'. The main content area is titled 'Rate a course you attended. One response per course, per program.' It contains a 'New feedback' form with a 'Training program' dropdown (Java Backend Bootcamp), a 'Course' dropdown (Spring Boot Fundamentals), a 'Rating' section with 5 stars (4 selected), and a 'Comments' text area with the text 'This is the testing'. A 'Submit feedback' button is at the bottom. To the right, an 'Already submitted' section shows a table of previously submitted feedback.

| COURSE                   | PROGRAM                | RATING |
|--------------------------|------------------------|--------|
| Spring Boot Fundamentals | Java Backend Bootcamp  | 5 ☆☆☆☆ |
| Core Java                | Java Backend Bootcamp  | 5 ☆☆☆☆ |
| MongoDB Essentials       | Data Engineering Track | 5 ☆☆☆☆ |

The screenshot shows the 'My Feedback' page in the Feedback Management system. The page has a sidebar with 'Overview', 'Dashboard', 'Courses', 'Training Programs', 'Submit Feedback', and 'My Feedback'. The main content area is titled 'My Feedback' and contains a table of submitted feedback.

| COURSE                   | PROGRAM                | RATING | COMMENTS                                    | SUBMITTED              | ACTIONS |
|--------------------------|------------------------|--------|---------------------------------------------|------------------------|---------|
| Spring Boot Fundamentals | Java Backend Bootcamp  | 5 ☆☆☆☆ | Clear explanations and great hands-on labs. | Jul 9, 2026, 6:05 AM   | Edit    |
| Core Java                | Java Backend Bootcamp  | 5 ☆☆☆☆ | The streams module was excellent.           | Jul 9, 2026, 6:05 AM   | Edit    |
| MongoDB Essentials       | Data Engineering Track | 5 ☆☆☆☆ | Best database course I have taken.          | Jul 9, 2026, 6:05 AM   | Edit    |
| MongoDB Essentials       | Java Backend Bootcamp  | 4 ☆☆☆☆ | This is the testing                         | Jul 14, 2026, 10:45 AM | Edit    |

Feedback Management

Alice Kumar  
PARTICIPANT  
Sign out

OVERVIEW

Dashboard

MANAGE

Courses

Training Programs

FEEDBACK

Submit Feedback

My Feedback

My Feedback

Everything you have submitted. You can revise your own responses at any time.

+ Submit feedback

| COURSE                   | PROGRAM                | RATING | COMMENTS                           | SUBMITTED              | ACTIONS                           |
|--------------------------|------------------------|--------|------------------------------------|------------------------|-----------------------------------|
| Spring Boot Fundamentals | Java Backend Bootcamp  | 5 ☆☆☆☆ | This is the testing Edit feedback  | Jul 9, 2026, 6:05 AM   | <div>Edited</div> <div>Edit</div> |
| Core Java                | Java Backend Bootcamp  | 5 ☆☆☆☆ | The streams module was excellent.  | Jul 9, 2026, 6:05 AM   | <div>Edit</div>                   |
| MongoDB Essentials       | Data Engineering Track | 5 ☆☆☆☆ | Best database course I have taken. | Jul 9, 2026, 6:05 AM   | <div>Edit</div>                   |
| MongoDB Essentials       | Java Backend Bootcamp  | 4 ☆☆☆☆ | This is the testing                | Jul 14, 2026, 10:45 AM | <div>Edit</div>                   |

## 5. Training Summary Report

Scenario: An admin views a summary across training programs showing enrolment, feedback count, response rate, defaulter count, and average rating.

Output:

Feedback Management

mayank  
ADMIN  
Sign out

OVERVIEW

Dashboard

MANAGE

Courses

Training Programs

Faculty

REPORTS

Feedback Reports

Defaulters

Training Summary

Course Analytics

Training Summary

Participation and feedback health across 2 programs.

Programs

2

Total enrolled

6

Feedback received

8

Overall response rate

83.3%

1 defaulters

Response rate by program

Java Backend ...

100%

Data Engineer...

50%

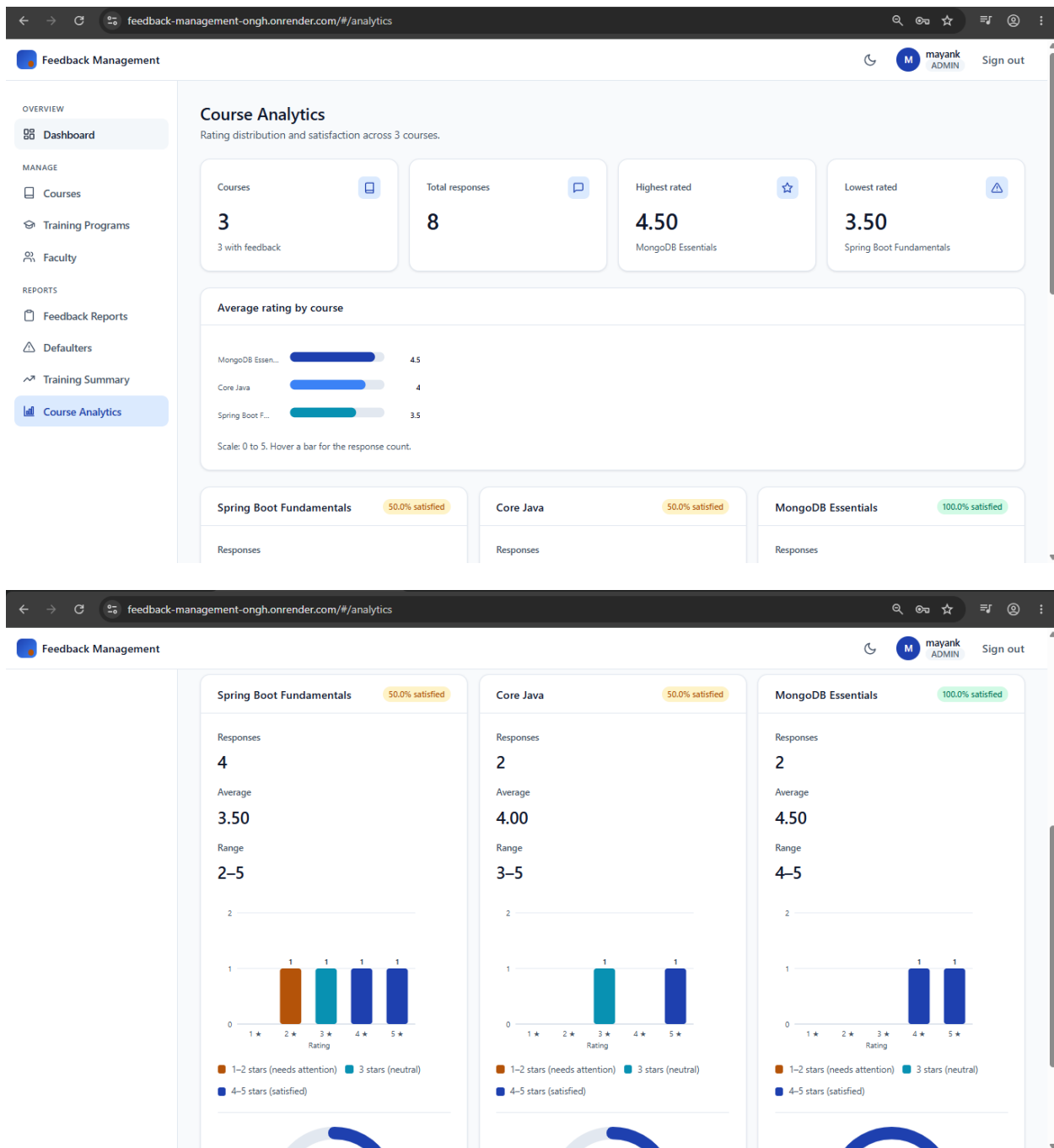
All programs

| PROGRAM               | STATUS  | SCHEDULE                   | ENROLLED | RESPONSES | DEFAULTERS | RESPONSE RATE | AVG RATING |
|-----------------------|---------|----------------------------|----------|-----------|------------|---------------|------------|
| Java Backend Bootcamp | ONGOING | Jun 1, 2026 → Aug 24, 2026 | 4 / 20   | 7         | 0          | 100.0%        | 3.7 ☆☆☆☆   |

## 6. Course Feedback Analytics

Scenario: An admin analyses feedback at the course level and the analytics are displayed correctly on the dashboard.

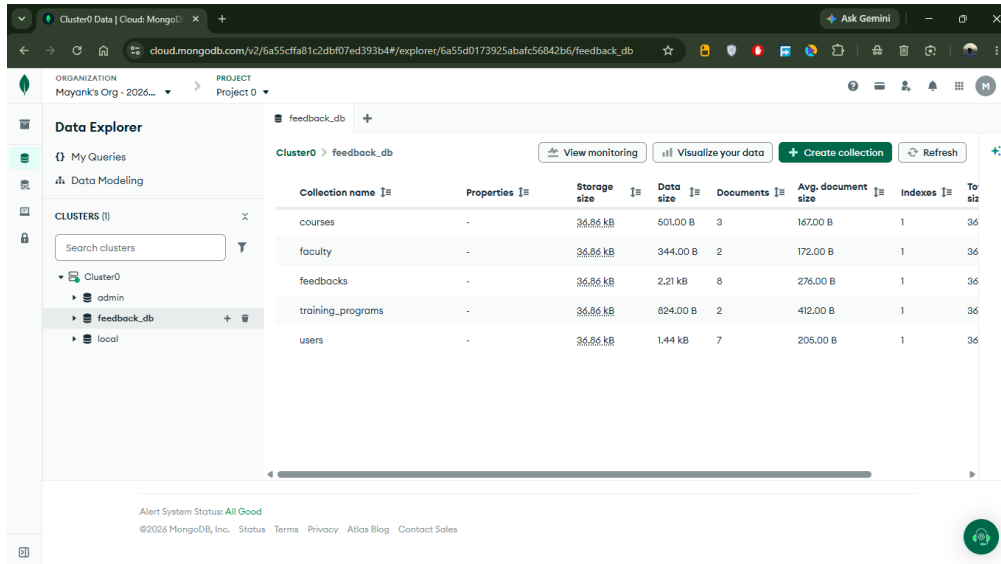
Output:



## 7. MongoDB Integration

Scenario: The application persists every collection: users, faculty, skills, courses, training programs, and feedback to MongoDB, with the connection string supplied by the MONGODB\_URI environment variable.

Output:



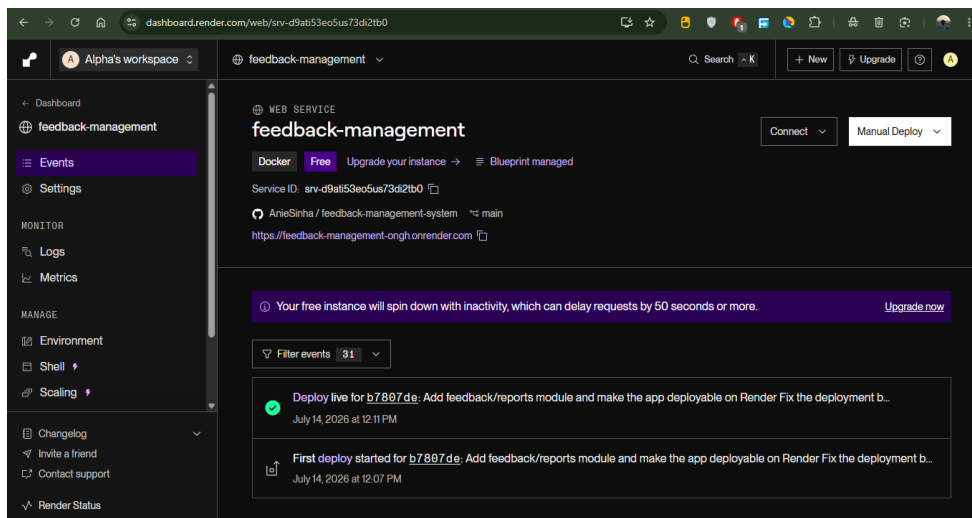
The screenshot shows the MongoDB Atlas Data Explorer interface. The left sidebar displays the 'Data Explorer' section with 'My Queries' and 'Data Modeling' options. The 'CLUSTERS' list on the left shows 'Cluster0' with sub-clusters 'admin', 'feedback\_db', and 'local'. The 'feedback\_db' cluster is selected, and the main panel shows a table of collections. The table has columns: Collection name, Properties, Storage size, Data size, Documents, Avg. document size, Indexes, and To size. The collections listed are: courses, faculty, feedbacks, training\_programs, and users.

| Collection name   | Properties | Storage size | Data size | Documents | Avg. document size | Indexes | To size |
|-------------------|------------|--------------|-----------|-----------|--------------------|---------|---------|
| courses           | -          | 36.86 kB     | 501.00 B  | 3         | 167.00 B           | 1       | 36      |
| faculty           | -          | 36.86 kB     | 344.00 B  | 2         | 172.00 B           | 1       | 36      |
| feedbacks         | -          | 36.86 kB     | 2.21 kB   | 8         | 276.00 B           | 1       | 36      |
| training_programs | -          | 36.86 kB     | 824.00 B  | 2         | 412.00 B           | 1       | 36      |
| users             | -          | 36.86 kB     | 1.44 kB   | 7         | 205.00 B           | 1       | 36      |

## 8. Deployment on Render

Scenario: The application is containerized with a Docker image and deployed as a Render web service backed by MongoDB Atlas. Secrets are injected as environment variables and the service is live and reachable on a public URL: [feedback-management-ongh.onrender.com](https://feedback-management-ongh.onrender.com)

Output:



The screenshot shows the Render dashboard for a web service named 'feedback-management'. The service is running on a 'Free' Docker instance. The dashboard displays the service ID, the user 'ArieSinha', and the public URL 'https://feedback-management-ongh.onrender.com'. A warning message states: 'Your free instance will spin down with inactivity, which can delay requests by 50 seconds or more.' Below this, there are two deployment events: 'Deploy live for b7807de: Add feedback/reports module and make the app deployable on Render Fix the deployment b...' and 'First deploy started for b7807de: Add feedback/reports module and make the app deployable on Render Fix the deployment b...'. The left sidebar shows navigation options like 'Dashboard', 'Events', 'Settings', 'Logs', 'Metrics', 'Environment', 'Shell', 'Scaling', 'Changelog', 'Invite a friend', 'Contact support', and 'Render Status'.

## **Objective**

Capture participant feedback reliably, turn it into reports that show response rates and sentiment, and deploy the whole system to a public URL.

## **Key Analysis**

- Build the feedback entity, repository, and submission API
- Prevent duplicate feedback and allow edits before the deadline
- Generate feedback reports for admins and coordinators
- Build defaulter tracking for participants who have not responded
- Aggregate training summary and course-level analytics
- Connect the application to MongoDB Atlas via environment configuration
- Containerise the application and deploy it on Render

## **Outcomes**

- Participants can submit and revise feedback without duplication
- Admins and coordinators can see response rates and average ratings
- Defaulters are identified so follow-up can be targeted
- All data persists in a managed MongoDB Atlas cluster
- The application runs live on Render with no secrets in the repository